

ORF307: Optimization

Spring 2024 2nd Midterm Exam Solutions

April 16, 11:00am – 12:20pm

Exam Rules

1. Please write out and sign the following pledge on the solution booklet: “I pledge my honor that I have not violated the Honor Code or the rules specified by the instructor during this exam.”
2. You are allowed a single sheet of paper, double-sided, hand-written or typed.
3. You cannot communicate with anyone during the exam.
4. All questions should be answered in the solution booklet. Write your name on every page that you use to write the solutions. Start each problem on a separate page. Answering multiple subparts of the same problem on the same page is fine. We will **not** grade anything written on the exam questions; we will only grade what you write in the solution booklet.
5. There are 3 problems worth a total of 100 points.
6. At the end of the exam, please turn in both your solution booklet and the exam questions.

1. *Problem 1 (35 points). The Juice Bar's Blending Dilemma.*

John owns a world-famous juice bar and produces his juice for the day in bulk. His shop's popularity means that John is always able to sell all of the juice that he makes. He produces two different juices: Tropical Sunrise and Green Detox and needs your help to determine how many gallons of each to make.

- Each gallon of Tropical Sunrise results in a profit of \$20 and requires 4 pounds of fruit and 2 pounds of vegetables.
- Each gallon of Green Detox results in a profit of \$10 and requires 3 pounds of fruit and 1 pound of vegetables.
- John's suppliers can provide him with **at most** 200 pounds of fruit and **at most** 120 pounds of vegetables.

Let x_1 be the number of gallons of Tropical Sunrise and x_2 be the number of gallons of Green Detox that John produces. Clearly, he cannot make a negative amount of either juice.

- (5 points) Write a linear program to maximize John's profit subject to the above constraints.
- (10 points) Convert the LP from part (a) to standard form and construct a basic feasible solution with $(x_1, x_2) = (0, 0)$. What is the basis B ?
- (10 points) Carry out one iteration of the simplex method from the initial basic feasible solution and basis you constructed in part (b). Use Bland's Rule if necessary. Specifically, report the values of the reduced costs vector $\bar{c}$, the basic direction d , the step size θ , the new basic feasible solution x^{new} , and the new basis B^{new} .
- (10 points) With the new basic feasible solution and basis from the end of part (c), compute the reduced costs vector $\bar{c}$. What would you do if you wanted to complete the simplex method?

Solution.

- (a) The LP can be written as:

$$\begin{aligned} \text{maximize} \quad & 20x_1 + 10x_2 \\ \text{subject to} \quad & 4x_1 + 3x_2 \leq 200 \\ & 2x_1 + x_2 \leq 120 \\ & x_1, x_2 \geq 0. \end{aligned} \tag{1}$$

(b) In standard form, problem (1) can be written as:

$$\begin{aligned}
& \text{minimize} && -20x_1 - 10x_2 \\
& \text{subject to} && 4x_1 + 3x_2 + s_1 = 200 \\
& && 2x_1 + x_2 + s_2 = 120 \\
& && x_1, x_2, s_1, s_2 \geq 0.
\end{aligned} \tag{2}$$

The basic feasible solution satisfying $(x_1, x_2) = (0, 0)$ is given by

$$x = (x_1, x_2, s_1, s_2) = (0, 0, 200, 120),$$

with $B = \{3, 4\}$.

(c) From the standard form in (2), we have

$$\begin{aligned}
c &= (-20, -10, 0, 0) \\
A &= \begin{bmatrix} 4 & 3 & 1 & 0 \\ 2 & 1 & 0 & 1 \end{bmatrix} \\
b &= (200, 120).
\end{aligned}$$

For the reduced costs, we know $\bar{c} = c - A^T p$ with $A_B^T p = c_B$. With the given x, B ,

$$A_B = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad c_B = (0, 0) \implies p = (0, 0), \quad \bar{c} = c = (-20, -10, 0, 0).$$

Bland's Rule tells us to add $j = 1$ to the basis. The basic direction satisfies $d_1 = 1$, $A_B d_B = -A_1$, and 0 elsewhere. Since $A_1 = (4, 2)$, we have $d_B = (-4, -2)$, so $d = (1, 0, -4, -2)$. For the step size,

$$\theta = \min\{-200/(-4), -120/(-2)\} = 50.$$

So,

$$x^{\text{new}} = x + \theta d = (0, 0, 200, 120) + 50(1, 0, -4, -2) = (50, 0, 0, 20),$$

which means $i = 3$ leaves the basis so $B^{\text{new}} = \{1, 4\}$.

(d) With the new basis and iterate, we have

$$A_B = \begin{bmatrix} 4 & 0 \\ 2 & 1 \end{bmatrix}, \quad c_B = \begin{bmatrix} -20 \\ 0 \end{bmatrix} \implies p = A_B^{-T} c_B = \frac{1}{4} \begin{bmatrix} 1 & -2 \\ 0 & 4 \end{bmatrix} \begin{bmatrix} -20 \\ 0 \end{bmatrix} = \begin{bmatrix} -5 \\ 0 \end{bmatrix}.$$

So, the reduced costs are $\bar{c} = c - A^T p = (0, 5, 5, 0)$. Since $\bar{c} \geq 0$, the solution $x = (50, 0, 0, 20)$ is optimal and we terminate the simplex algorithm.

2. Problem 2 (35 points). Saturday Time Allocation.

It's Saturday afternoon and you want to split your time between two activities: going to the gym, and reading a new book. Each hour at the gym gives you one unit of happiness, and each hour of reading also gives you one unit of happiness. You want to maximize your total happiness, but there is a limit on how much you can do.

- Each hour at the gym uses up 3 units of stamina, while each hour reading recovers 1 unit of stamina. You want to end the afternoon with **at most** 1 unit of stamina. (A negative stamina value is allowed - you will just need to recover at night.)
- Each hour at the gym increases the level of your hunger by 2, and each hour reading increases it by 1. Your hunger level can be **at most** 6 before you end these activities and go for dinner instead.

Let x_1 be the number of hours spent at the gym, and x_2 be the number of hours spent reading. You cannot spend a negative number of hours on either activity.

- (a) (5 points) Write a linear program to maximize your happiness, subject to the above constraints.
- (b) (5 points) Write the problem from part (a) as an LP of the following form:

$$\begin{aligned} & \text{minimize} && c^T x \\ & \text{subject to} && Ax \leq b. \end{aligned} \tag{1}$$

- (c) (5 points) Write down the dual problem of the LP you derived in part (b). Expand your answer with the problem data instead of just writing the general dual formulation.
- (d) (10 points) Show that feasible $x = (1, 4)$ is optimal.
Hint: Use complementary slackness.
- (e) (5 points) You realize you like going to the gym more than reading a book. Each hour at the gym now gives you $(1 + \epsilon)$ units of happiness. How does the optimal value p^* change as a function of ϵ around $\epsilon = 0$? Give an explicit expression for $p^*(\epsilon)$.
- (f) (5 points) Suppose your eyes tire if you read too long, so you can only read up to 3 hours in one afternoon. What is the change to the primal problem (not including the changes made in part (e))? Give a **dual feasible** solution to the new problem.

Solution.

(a) The optimization problem is

$$\begin{aligned}
 & \text{maximize} && x_1 + x_2 \\
 & \text{subject to} && -3x_1 + x_2 \leq 1 \\
 & && 2x_1 + x_2 \leq 6 \\
 & && x_1 \geq 0 \\
 & && x_2 \geq 0.
 \end{aligned} \tag{2}$$

(b) Converting the above to inequality form, we have

$$c = \begin{bmatrix} -1 \\ -1 \end{bmatrix}, \quad A = \begin{bmatrix} -3 & 1 \\ 2 & 1 \\ -1 & 0 \\ 0 & -1 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 6 \\ 0 \\ 0 \end{bmatrix}.$$

(c) The dual problem is given by

$$\begin{aligned}
 & \text{maximize} && -y_1 - 6y_2, \\
 & \text{subject to} && -3y_1 + 2y_2 - y_3 - 1 = 0, \\
 & && y_1 + y_2 - y_4 - 1 = 0, \\
 & && y \geq 0.
 \end{aligned} \tag{3}$$

(d) The first and second constraints are active at x . Thus, we consider $y = (y_1, y_2, 0, 0)$. By the constraints in (3), we have

$$\begin{bmatrix} -3 & 2 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \end{bmatrix},$$

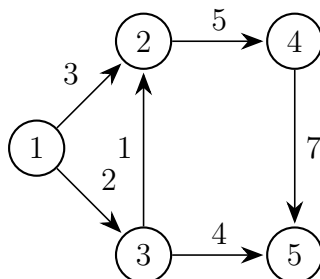
which implies $y_1 = 1/5$ and $y_2 = 4/5$. Since all the elements of $y = (1/5, 4/5, 0, 0)$ are non-negative, we obtain a feasible solution for the dual problem (3). The duality gap is 0, with $p^* = d^* = -5$. We conclude that x is primal optimal.

(e) Refer to HW7 Q1. The equation is $p^*(\epsilon) = p^*(0) - \epsilon x_1^* = -5 - \epsilon$.

(f) We add the constraint $x_2 \leq 3$ to the primal problem. A dual feasible solution is $(y^*, 0)$.

3. Problem 3 (30 points). Shortest Path Problem.

Consider a directed graph shown below representing a transportation network. The network has cost vector $c = (3, 2, 5, 1, 4, 7)$ and these arc costs are shown on the diagram.



The given cost vector corresponds to the following arc ordering (start, end):

Arc index	1	2	3	4	5	6
Connected nodes	(1, 2)	(1, 3)	(2, 4)	(3, 2)	(3, 5)	(4, 5)

The goal is to find the shortest path from node 1 to node 5, where the ‘lengths’ of each arc come from c .

- (5 points) Write down the arc-node incidence matrix for the network with the given arc ordering.
- (5 points) Formulate the shortest path problem as a minimum cost network flow linear program.
- (10 points) Derive the dual of the linear program from part (b).
- (10 points) Given that

$$x^* = (0, 1, 0, 0, 1, 0)$$

is an optimal solution to our problem, use complementary slackness to infer which dual variables are zero or which dual constraints are tight.

Solution:

- The arc-node incidence matrix A is given by:

$$A = \begin{bmatrix} 1 & 1 & 0 & 0 & 0 & 0 \\ -1 & 0 & 1 & -1 & 0 & 0 \\ 0 & -1 & 0 & 1 & 1 & 0 \\ 0 & 0 & -1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & -1 & -1 \end{bmatrix}$$

- (b) We encode the shortest path problem by sending one unit of flow from node 1 to node 5. That is, the supply vector e is given by

$$e = (1, 0, 0, 0, -1).$$

Each arc is either included or not included on the shortest path, which are represented by the binary constraints $x \in \{0, 1\}^n$, and by total unimodularity of A we relax the binary constraints to derive the final LP:

$$\begin{aligned} & \text{minimize} && c^T x \\ & \text{subject to} && Ax = e \\ & && 0 \leq x \leq \mathbf{1}. \end{aligned} \tag{1}$$

- (c) To take the dual of (1), we rewrite the LP as:

$$\begin{aligned} & \text{minimize} && c^T x \\ & \text{subject to} && Ax = e \\ & && -x \leq 0 \\ & && x \leq \mathbf{1}. \end{aligned} \tag{2}$$

By assigning dual variables y, z, w , respectively to the sets of constraints, we derive the dual as:

$$\begin{aligned} & \text{maximize} && -e^T y - \mathbf{1}^T w \\ & \text{subject to} && A^T y - z + w + c = 0 \\ & && z, w \geq 0. \end{aligned} \tag{3}$$

This was sufficient for the answer, but note that z is just a slack variable, so the following is also correct:

$$\begin{aligned} & \text{maximize} && -e^T y - \mathbf{1}^T w \\ & \text{subject to} && A^T y + w + c \geq 0 \\ & && w \geq 0. \end{aligned} \tag{4}$$

- (d) If the dual was of the form in (3), then by writing

$$\begin{aligned} z &= (z_1, z_2, z_3, z_4, z_5, z_6) \\ w &= (w_1, w_2, w_3, w_4, w_5, w_6), \end{aligned}$$

complementary slackness tells us that

$$z_2^* = z_5^* = w_1^* = w_3^* = w_4^* = w_6^* = 0.$$

If the dual was of the form in (4), then the same result still holds for w , and instead of the z vector, we know that the tight dual inequalities are given by

$$(A^T y + w + c)_j = 0,$$

for $j = 2, 5$.